# SAMPLE

## Great Observatories: XMM-Newton

Europe's X-Ray Eye

Aira Astralis

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# Introduction

On the night of the tenth of December, 1999, an Ariane 5 rocket rose from the Guiana coast carrying the heaviest science satellite Europe had ever built, and the mission christened for Isaac Newton went to work. This book is the story of that observatory: why X-rays from the sky can only be caught from above the atmosphere, how three nestings of gold-coated mirrors grew into the largest X-ray collecting surface ever flown, what the machine has seen in the quarter century since, and why a telescope named for a seventeenth-century Englishman has spent the twenty-first century writing the book on matter at the extremes. The chapters follow the arc of the series: the mission's conception and the sisterhood of instruments it belongs to, the engineering of the spacecraft and its instruments, the physics of the hot universe it observes, the choreography of its long elliptical orbits, and the discoveries, challenges, and future of a mission that has outlived its design life more than ten times over.

The story deserves a word on method before it begins. X-ray astronomy is the astronomy of violence and heat: gas at millions of degrees, black holes feeding, neutron stars spinning, the wreckage of exploded stars, the vast atmospheres of galaxy clusters. Every chapter of the sky's hot side that the reader has met in a news item over the past twenty-five years has, in some part, been read from this machine's data, and the chapters that follow are written to let the reader understand how a spacecraft four tonnes in mass, on an orbit that swings it a hundred thousand kilometers out every two days, turns X-rays into knowledge. The plain-language promise of the series holds here as everywhere: no physics degree is required, the numbers are given where they earn their place, and the technical terms arrive when they are needed, introduced in words rather than equations.

The chapters also keep, in the series' practice, the human ledger: the rocket-era founders who proposed the machine, the mirror-builders who electroformed its gold, the control-room teams who have held it on its loop for a quarter of the century, and the archive-miners who will use it after everyone now flying has retired, a cast the reader will meet at every stage. One caution, familiar to readers of the series, and one pleasure. The caution: an observatory still operating is a moving target, and the discovery record summarized in the middle chapters is a snapshot of a mission whose archive grows nightly; the reader should expect the machine to outlast this book's numbers as it has outlasted every number ever set for it.

The pleasure: of all the instruments in this series, this is the marathon runner, the one that has simply refused to end, and its story is therefore the clearest

case the series can offer of how a spacecraft becomes an institution, with schedules, careers, and a worldwide community of users organized around a machine that was designed for two years of life and has given a quarter of a century so far.

The observatory's genesis also has a human chapter the reader should meet, because the machine was proposed by the same generation that founded the field. The scientists who wrote the 1980s mission studies had begun their careers in the rocket era, launching instruments on flights measured in minutes, and their advocacy for a cornerstone-class observatory, the biggest thing their agencies could fly, was the argument that a field this young deserved an instrument this large. The argument won on the science, but it also won on the record the early satellites had already set: the first imaging mission had shown what focus did for a field of point sources, and the community's case was simply that a machine several times larger, with spectroscopy and timing built in, would do for the faint and fast sky what the earlier one had done for the bright. The reader should hold this lineage in mind through every chapter that follows: the observatory's data are the rocket era's inheritance, spent at scale.

A word on the machine's place in the wider cast of this series. The reader who has toured the other observatory volumes will find this one the sibling with the longest commute: unlike the mountain-top telescopes and the desert arrays of the companion books, this machine's workplace is emptier than any mountaineer's, a stretch of near-space threaded by a two-day loop, worked by radio from two continents and visited by no one since launch. The emptiness is the instrument's choice, for the X-ray sky's demand is not darkness or dryness but altitude itself, the total removal of the absorbing air, and the machine's design spends everything else, the launch mass, the schedule's rhythm, the career of its ground team, to buy that removal. The reader will meet the consequences of the purchase throughout the chapters: an observatory that never waits for night, never bows to weather, and never, in a quarter century of service, has been touched by a human hand.

The reader should also know how the mission's early years validated the builders' bets, because the validation is the discovery chapter's proper close. The first observing cycles established the machine's signature strengths in practice: exposures that caught whole fields of sources in one stare, spectra that resolved the hot gas the gratings were built for, timing that held the fast sources steady across the two-day loop, and an optical monitor that made every X-ray pointing a two-band observation. The early science also set the pattern of the surprise: the machine's wide grasp caught in its first years what no one had aimed it at, the flare in a field meant for a cluster, the source at

the field's edge that the follow-up found to be a quasar, and the archive, even then, was already becoming the co-discoverer it would be for the next two decades. The mission's designers had promised grasp, patience, and simultaneity; the early years delivered them, and the quarter century since has been the compounding.

The launch campaign itself, as an engineering event, closes the discovery chapter's account properly. The spacecraft crossed the Atlantic by sea to the spaceport, was mated to the new launcher under the schedules of the fleet's first operational year, and rode the debut of Europe's heavy-lift era into the sky, a choice of timing that put the mission's mass and the launcher's promise on the same wager; the wager paid, the deployment was clean, and the commissioning that followed, the months of first light, calibration, and checkout, ended with a machine performing to its mirror-builders' specifications. The reader should note what the commissioning proved, for it set the mission's character for the decades: the observatory arrived in orbit ready, its instruments aligned, its calibrations begun, and its first scientific observations delivered to the archive within weeks of the launch, a working observatory from the start rather than a testbed that slowly became one.

# Chapter 1: The Discovery of XMM-Newton

The discovery chapter of a space observatory's story is properly the story of how the sky's X-rays were found at all, and that story begins at the top of the atmosphere. X-rays from space cannot be observed from the ground: the atmosphere absorbs them utterly, a mercy for the biosphere and an obstacle for the astronomy, and the proof that the sky emits them at all had to wait for the rocket era. In 1949, instruments carried aloft by a captured rocket detected X-rays from the Sun; in 1962, a sounding rocket flown by a small team searching for lunar X-rays found, instead, a source in the constellation Scorpius, the first cosmic X-ray source beyond the solar system, and the field was born.

The early decades were spent with short rocket flights and then small satellites, each observing a handful of sources, and the lesson of those decades was that the X-ray sky was not a curiosity but a universe: thousands of sources, then tens of thousands, and behind them the hot matter that the optical sky could not show.

By the 1970s the field's ambition had a shape, and the shape was a large telescope. X-rays, it had been found, reflect only at grazing incidence, skimming off a mirror's surface at fractions of a degree the way a stone skips off water, so an X-ray telescope is not a dish but a barrel: nested rings of polished metal, each ring catching the light at its own grazing angle, all focused down the barrel together. The first imaging X-ray observatory, the Einstein satellite of 1978, proved the barrel design in orbit, and its European successors-in-waiting drew the obvious conclusion: the next machine should be the largest nest of mirrors the launchers of the day could carry.

The European Space Agency's planning documents of the 1980s, worked through with the community's high-energy astrophysicists, settled on a mission of three large mirror modules and a battery of instruments, and the project that emerged, the X-ray Multi-Mirror Mission, was approved as one of the cornerstones of the agency's science program, the largest and most expensive science satellite Europe had yet undertaken.

The mission's naming belongs in the chapter too, because the machine carries a name whose origins the reader should know honestly. The acronym describes the hardware, a multi-mirror X-ray telescope; the surname honors the physicist whose prism first split sunlight and whose mechanics first described the orbits the spacecraft would later study, and the mission was formally titled with both, the X-ray Multi-Mirror Mission, named for Newton. The observatory is frequently and wrongly imagined as a biography in orbit; the

chapters that follow give the name its proper weight and no more, for the machine's subject is not the man but the hot universe his optics and mechanics made measurable.

The satellite was built in the 1990s across the agency's member states, with the instruments contributed by consortia drawn from the same nations and from the United States, whose space agency joined the mission as a partner, and the completed spacecraft, nearly four tonnes with its solar wings and its telescope tube, was shipped across the Atlantic for launch from the Guiana spaceport.

The launch, on the tenth of December 1999, was the first operational flight of the Ariane 5, Europe's new heavy launcher, and the deployment into orbit began one of the longest careers in the history of space science. The spacecraft was placed on a highly elliptical path designed for long, uninterrupted observations, and the first light images followed within weeks, showing the machine's grasp: a field of sources in a single exposure, spectra of hot gas, and the first X-ray view of the quality the mirror builders had promised. The design life was two years, with the consumables sized for a decade of operations, and the mission's scientists have since rewritten that arithmetic with margins that the planning documents never dared: a quarter of a century after launch the machine is still observing, and the chapters that follow describe how it has done it, what it has found, and what it is still owed the chance to see.

A note on the machine's daily temperament belongs here too, for understanding a telescope includes understanding its weather. The spacecraft observes through an environment no ground-based observer knows: the particle radiation of the belts, the flares and storms the Sun sends across the years, the micro-punctuations of its own aging electronics, and the Earth's occlusion of targets on the low passes of its loop. The machine's observation plans are therefore negotiated with its environment, scheduling the deep exposures in the high clean stretches, shielding the detectors on the low dirty ones, and modeling, exposure by exposure, the background the environment imposes. An X-ray astronomer's weather is radiation, and the reader who understands that single fact understands the shape of every schedule the mission has ever published.

Understanding the machine also means understanding its economy, for the observatory is a marketplace as well as an instrument. Its time is the commodity: proposed for, competed for, oversubscribed by a factor the mission's reports publish with every call; its currency is the observing orbit, the unit the schedule is denominated in; and its dividend is the proprietary

interval, the protected year in which the proposers alone may mine their data, after which the observation joins the public archive and the common wealth. The mission's economy has held for a quarter century without a devaluation of its rules, and the field's trust in it, expressed in the cycle-over-cycle rise of the archive's citation share, is the soundest measure of the market the machine keeps. The reader who has followed the series knows this economy by other names from other books; here it is worth naming plainly, because the machine's time is among the scarcest commodities the astronomical world trades in.

Understanding also requires the machine's family tree in the sky, brief and clear. The mission's sister observatory, launched the same year, holds the sharpness record of the pair and the decade; the harder X-ray telescope of the 2010s added the band above the machine's reach; the survey mission of the 2010s mapped the whole sky the machine stares into; and the timing and burst missions of the era added the cadences the pointed observatories cannot keep. The family shares the sky through the alert networks and the coordinated campaigns the earlier chapters described, and the machine's role in the family, agreed rather than assigned, is the deep spectrum and the long stare, the reference observation the others' discoveries are measured against. The reader who wants one sentence of understanding should take this: the hot sky is watched by a fleet, and this machine is the fleet's oldest serving member, the one the newcomers still call for the standard measurement.

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# ABOUT THE AUTHOR

Aira Astralis is a science writer specializing in astronomy and space exploration. Her work spans the full breadth of humanity's relationship with the cosmos -- from the earliest naked-eye observations to the engineering marvels of modern space missions.

Her books are built on a foundation of primary sources: peer-reviewed research, mission archives, and direct engagement with the scientific community. She writes for readers who want to understand not just what we know about the universe, but how we came to know it.

She believes that the most important frontier is not out there but in the gap between what scientists discover and what the public understands. Her books exist to close that gap.